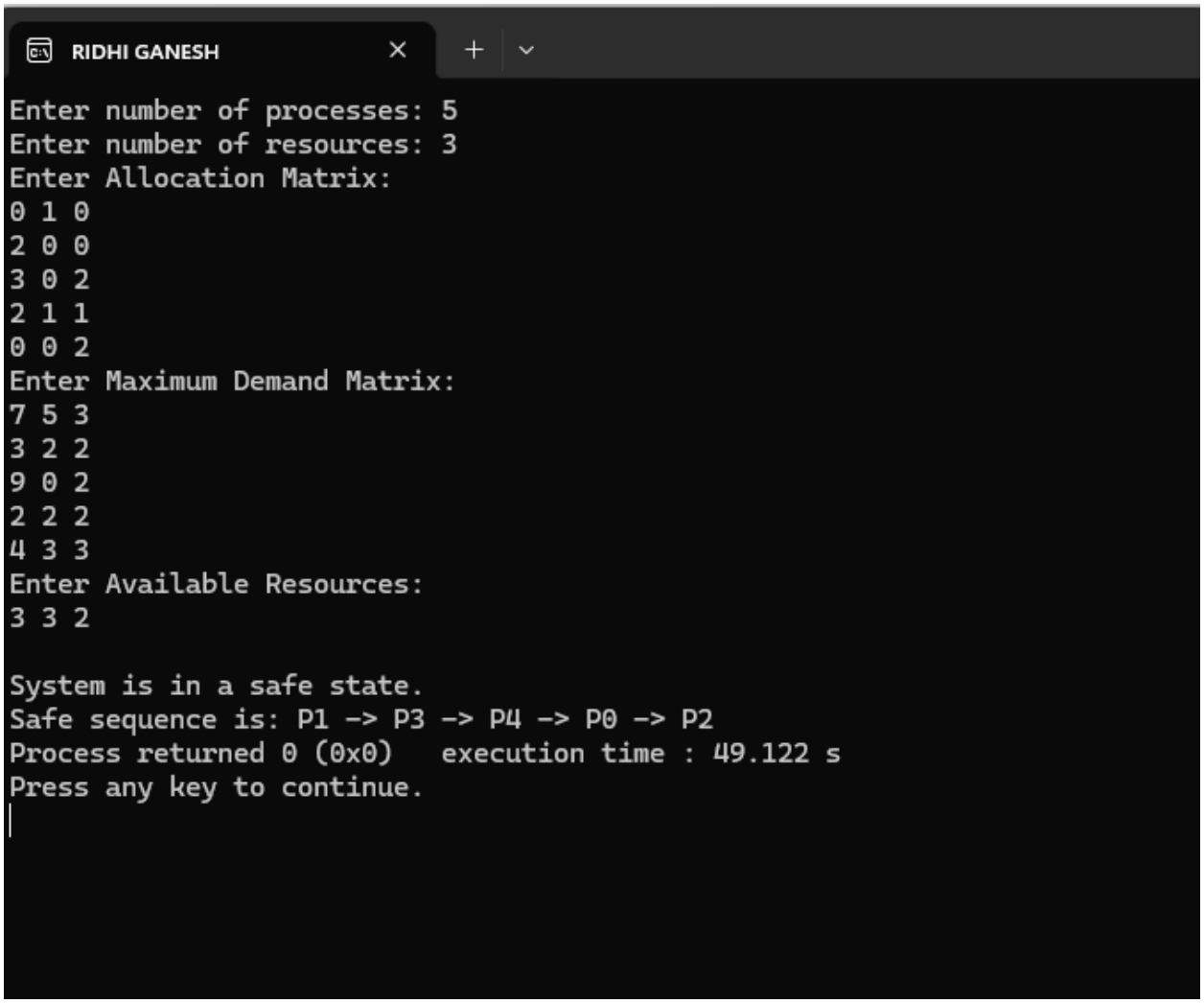
5A)SAFETY ALGORITHM

```
#include <stdio.h>

int main() {
    int n, m, i, j, k;

    printf("Enter number of processes: ");
    scanf("%d", &n);

    printf("Enter number of resources: ");
    scanf("%d", &m);

    int alloc[n][m], max[n][m], need[n][m];
    int avail[m], finish[n], safe[n];

    // Allocation Matrix
    printf("Enter Allocation Matrix:\n");
    for(i = 0; i < n; i++) {
        for(j = 0; j < m; j++) {
            scanf("%d", &alloc[i][j]);
        }
    }

    // Maximum Matrix
    printf("Enter Maximum Demand Matrix:\n");
    for(i = 0; i < n; i++) {
        for(j = 0; j < m; j++) {
            scanf("%d", &max[i][j]);
        }
    }

    // Available Resources
    printf("Enter Available Resources:\n");
    for(j = 0; j < m; j++) {
        scanf("%d", &avail[j]);
    }

    // Need Matrix = Max - Allocation
    for(i = 0; i < n; i++) {
        for(j = 0; j < m; j++) {
            need[i][j] = max[i][j] - alloc[i][j];
        }
    }
```

```
// Initialize finish as false
for(i = 0; i < n; i++) {
   finish[i] = 0;
}

int count = 0;

while(count < n) {
   int found = 0;

   for(i = 0; i < n; i++) {

      if(finish[i] == 0) {

         int possible = 1;

         // Check Need <= Available
         for(j = 0; j < m; j++) {
            if(need[i][j] > avail[j]) {
               possible = 0;
               break;
            }
         }

         // Process can execute
         if(possible) {

            for(k = 0; k < m; k++) {
               avail[k] += alloc[i][k];
            }

            safe[count] = i;
            finish[i] = 1;
            count++;
            found = 1;
         }
      }
   }

   // Unsafe state
   if(found == 0)
      break;
}
```

```
    // Check safe state
    if(count == n) {
        printf("\nSystem is in a safe state.\n");
        printf("Safe sequence is: ");

        for(i = 0; i < n; i++) {
            printf("P%d", safe[i]);

            if(i != n - 1)
                printf(" -> ");
        }
    }
    else {
        printf("\nSystem is NOT in a safe state.\n");
    }

    return 0;
}
```



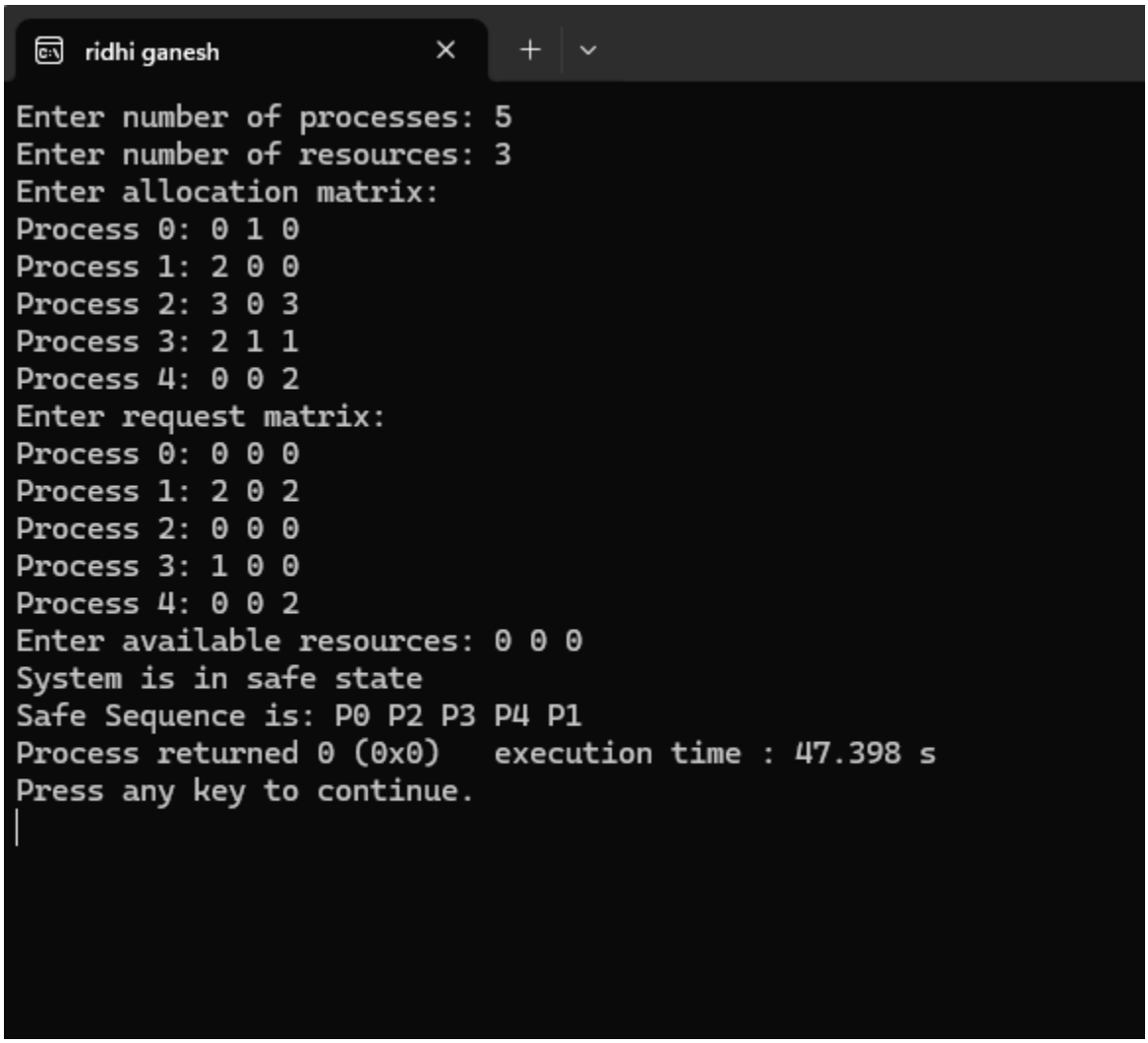
5B) DEADLOCK DETECTION

```
#include <stdio.h>

int main() {

    int n, m;
    int i, j;

    printf("Enter number of processes: ");
    scanf("%d", &n);

    printf("Enter number of resources: ");
```

```
scanf("%d", &m);

int alloc[n][m];
int req[n][m];
int avail[m];
int finish[n];
int safe[n];

// Allocation matrix
printf("Enter allocation matrix:\n");
for(i = 0; i < n; i++) {
    printf("Process %d: ", i);
    for(j = 0; j < m; j++) {
        scanf("%d", &alloc[i][j]);
    }
}

// Request matrix
printf("Enter request matrix:\n");
for(i = 0; i < n; i++) {
    printf("Process %d: ", i);
    for(j = 0; j < m; j++) {
        scanf("%d", &req[i][j]);
    }
}

// Available resources
printf("Enter available resources: ");
for(i = 0; i < m; i++) {
    scanf("%d", &avail[i]);
}

// Initially all are unfinished
for(i = 0; i < n; i++) {
    finish[i] = 0;
}

int count = 0;

while(count < n) {

    int found = 0;

    for(i = 0; i < n; i++) {
```

```
        if(finish[i] == 0) {

            int possible = 1;

            // Check request <= available
            for(j = 0; j < m; j++) {

                if(req[i][j] > avail[j]) {
                    possible = 0;
                }
            }

            // If possible execute process
            if(possible == 1) {

                for(j = 0; j < m; j++) {
                    avail[j] = avail[j] + alloc[i][j];
                }

                safe[count] = i;
                count++;
                finish[i] = 1;
                found = 1;
            }
        }
    }

    // Unsafe state
    if(found == 0) {
        break;
    }
}

// Final Output
if(count == n) {

    printf("System is in safe state\n");

    printf("Safe Sequence is: ");

    for(i = 0; i < n; i++) {
        printf("P%d ", safe[i]);
    }
```

```
    }
    else {

        printf("System is NOT in safe state");
    }

    return 0;
}
```



1WA24CS229